

2026

CAB230 Assignment 2 Client Side



CAB230

Rentals API – Client Side Application

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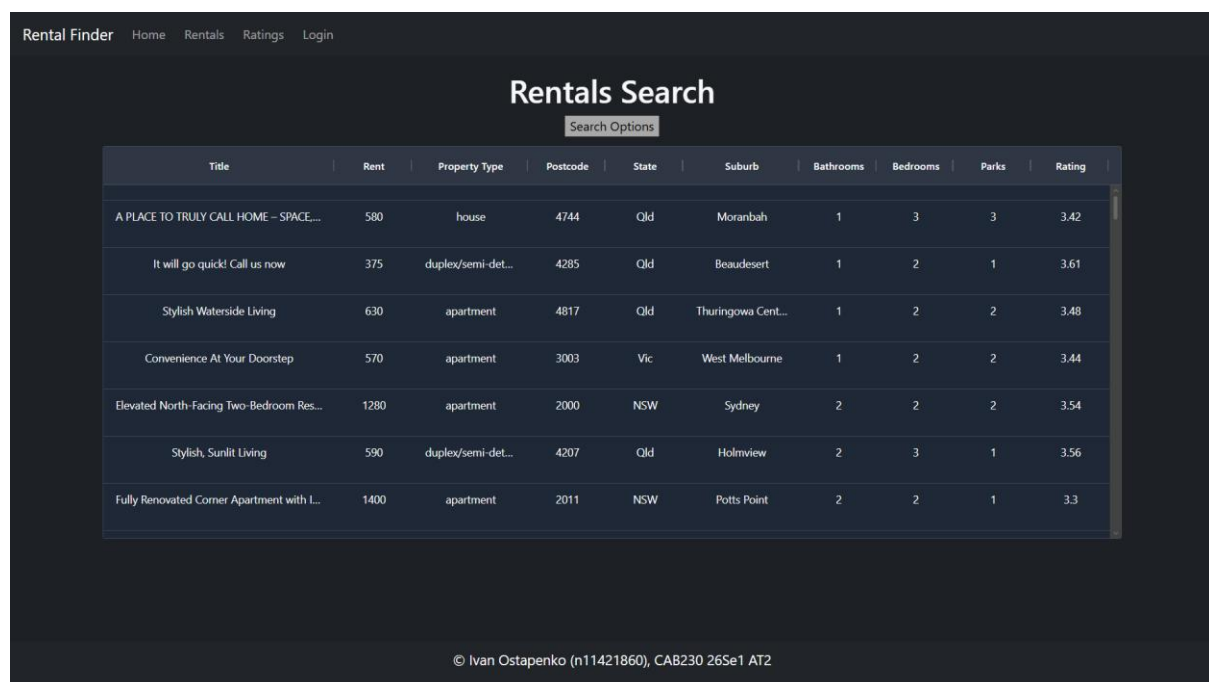
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Introduction

Purpose & description

The “Rental Search” web application is created to allow users to search and view data about rental properties across Australia. The information about properties is presented in a form of a table (*Screenshot 1*), and more detailed information about property may be accessed, including its location on a map. Properties may be filtered by various parameters, including: property title, state, postal code, suburb, rent, rating, etc., which updates the table contents accordingly. The application allows users to create an account to leave their ratings across various properties. When logged in, user may view all their ranked properties on a dedicated page. User can navigate through different pages using provided UI elements.



Rentals Search									
Search Options									
Title	Rent	Property Type	Postcode	State	Suburb	Bathrooms	Bedrooms	Parks	Rating
A PLACE TO TRULY CALL HOME – SPACE...	580	house	4744	Qld	Moranbah	1	3	3	3.42
It will go quick! Call us now	375	duplex/semi-det...	4285	Qld	Beaudesert	1	2	1	3.61
Stylish Waterside Living	630	apartment	4817	Qld	Thuringowa Cent...	1	2	2	3.48
Convenience At Your Doorstep	570	apartment	3003	Vic	West Melbourne	1	2	2	3.44
Elevated North-Facing Two-Bedroom Res...	1280	apartment	2000	NSW	Sydney	2	2	2	3.54
Stylish, Sunlit Living	590	duplex/semi-det...	4207	Qld	Holmview	2	3	1	3.56
Fully Renovated Corner Apartment with L...	1400	apartment	2011	NSW	Potts Point	2	2	1	3.3

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Screenshot 1. Rental data table

This is a single page application, in which react-router-dom library has been utilized to route between various components and handle public/protected routes for visitors and registered users. The data tables are rendered using Ag-grid-react library. In order to create a somewhat modern looking and adaptive UI, the Bootstrap v5 spacing utilities and CSS has been used, along with multiple style sheets written for the specific pages. A react-toastify library has been used for displaying informative pop ups upon successful POST requests to the backend.

Completeness and Limitations

Application successfully implements most of the expected functionality. The rental and rating tables implement an infinite row model, which provides better navigation experience for the user. Signup, login and logout are supported, and bearer token is used to access endpoints which require authorization. An individual rental page can be accessed by clicking on respective row in the table, which provides all property information, including location. On the same page user may rate the property or change existing rating. Rated properties are shown on the respective page, and application provides the means to easily visit different pages via navigation bar, which is rendered on every page of the application. However, the app may be lacking in proper error handling.

Use of End Points

</rentals/states>

This endpoint is used to retrieve an array of states that are used by the rental database. The retrieved data is being used to provide search by state functionality. As can be seen on *Screenshot 2* below, radiobuttons are used to select one state from the list.

</rentals/property-types>

This endpoint is used to retrieve an array of property types that are used by the rental database. The retrieved data is being used to provide search by property types functionality. As can be seen on *Screenshot 2* below, checkboxes are used to select one or more property type from the list.

</rentals/search>

This endpoint is being used in two major ways:

- Retrieve rental properties data and build a table with provided information. “data” section of the JSON provides property information, while “pagination” is used for pagination functionality, or, in this case, an infinite scrolling rendering, as can be seen on *Screenshot 1* above.
- To request filtered and/or sorted rental data by providing built search query in the fetch() request. The various filtering parameters may be entered by user using the “Search Options” form seen on *Screenshot 2* below.

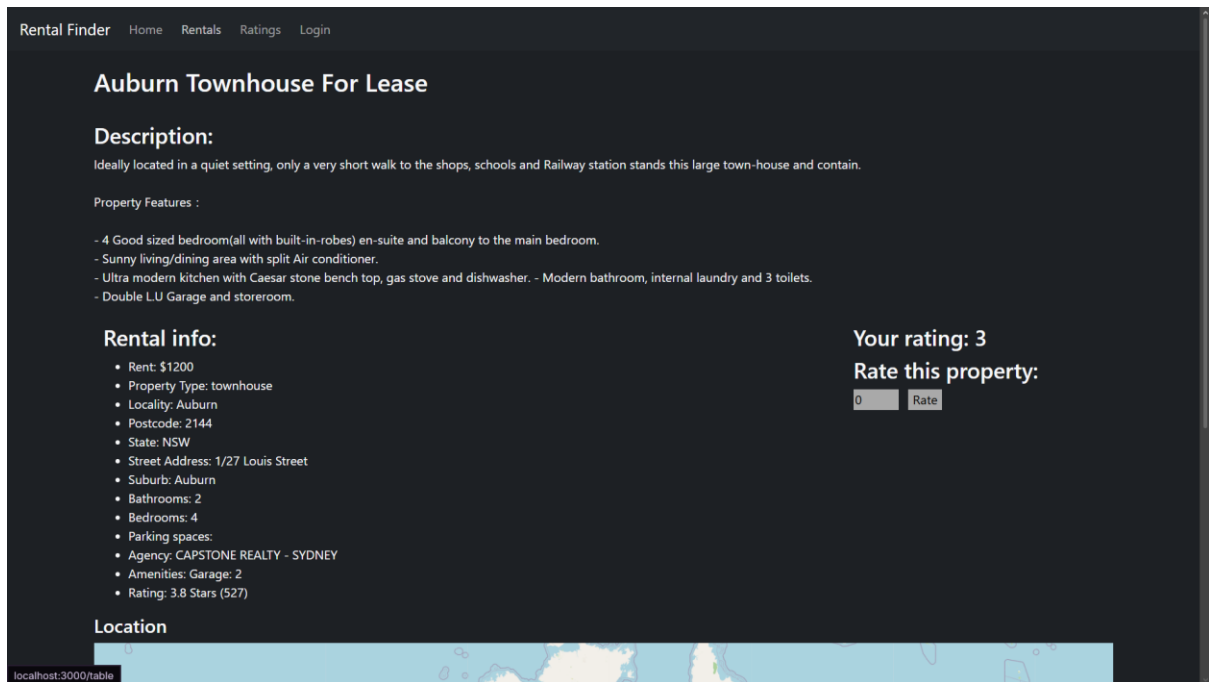
The screenshot shows a web application titled "Rental Finder" with a navigation bar containing "Home", "Rentals", "Ratings", and "Login". The main section is titled "Rentals Search" and contains a "Search Options" form. The form includes a "State" section with radio buttons for ACT, NSW, NT, Qld, SA, Tas, Vic, and WA. Below this is an "Advanced Search" section with checkboxes for various property types: acreage/semi-rural, apartment, duplex/semi-detached, flat, house, other, studio, terrace, townhouse, unit, and villa. To the right of these checkboxes are input fields for Suburb, Postcode, Min Rent (\$), Max Rent (\$), Min Bathrooms, Max Bathrooms, Min Bedrooms, Max Bedrooms, Min Parking, Max Parking, Min Rating, and Max Rating. At the bottom of the form are "Sort By" and "Sort Order" dropdown menus, both currently set to "None", and "Search" and "Reset" buttons. Below the form is a table displaying search results with the following columns: Title, Rent, Property Type, Postcode, State, Suburb, Bathrooms, Bedrooms, Parks, and Rating. The table contains three rows of data:

Title	Rent	Property Type	Postcode	State	Suburb	Bathrooms	Bedrooms	Parks	Rating
Auburn Townhouse For Lease	1200	townhouse	2144	NSW	Auburn	2	4	2	3.8
STUDIO ART-DECO APARTMENT	420	studio	2010	NSW	Darlinghurst	1	3	2	3.61
A PLACE TO TRULY CALL HOME - SPACE...	580	house	4744	Qld	Moranbah	1	3	3	3.45

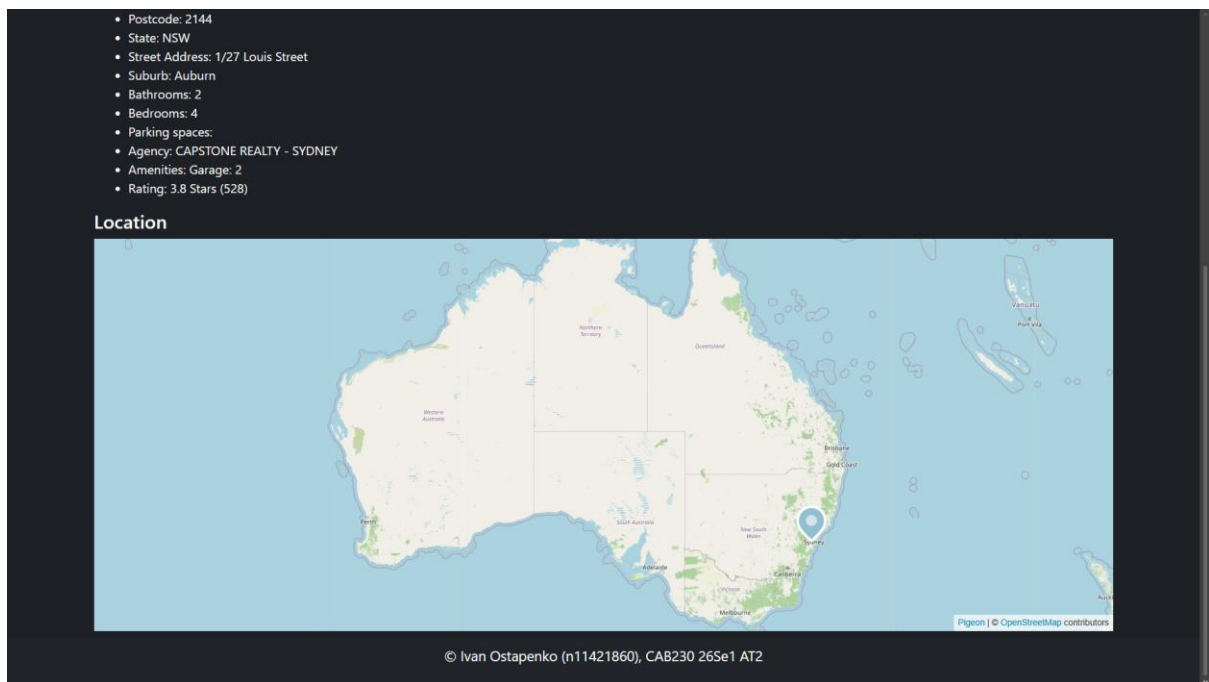
Screenshot 2. Rental Search Options

</rentals/{id}>

This endpoint provides all data about one rental property based on provided id in the request URL. This endpoint is called when user goes to specific property page when interacting with the respective rental or ratings table row. The description is shown separately from the rest of the values, and any HTML in the description is formatted accordingly. The map is used to visualize property location with marker indicating its exact position on the map.



Screenshot 3. Rental Page Information with user rating



Screenshot 4. Rental Page Map

/ratings

This endpoint requires a logged in user token to be accessed, so users that are not logged in are unable to access page with rental table. Similar to `/ratings/search` endpoint, JSON provides “data” with and “pagination”, which are similarly used to build a table. However, this endpoint provides no rental data besides ID, so `rental/id` endpoint needs to be called, and data merged with respective rating data to build a proper table. Datetime data needs to be formatted to be properly shown in the table.

Rental Finder Home Rentals Ratings Login									
Your Ratings									
Title	Rent	Property Type	Postcode	State	Suburb	Rating	Total Ratings	Your Ra...	Rated
Auburn Townhouse For Lease	1200	townhouse	2144	NSW	Auburn	3.8	531	3	5/3/2026, 7:10:35 PM
Stylish, Sunlit Living	590	duplex/semi-d...	4207	Qld	Holmview	3.57	108	5	5/3/2026, 6:44:15 PM
Ocean View in a Perfect Position!	520	unit	4802	Qld	Cannonvale	4.11	9	3	5/3/2026, 6:19:39 PM
Spacious Family Home with Brand...	820	house	4074	Qld	Mount Omma...	3.82	33	3	5/1/2026, 10:54:23 PM
STUDIO ART-DECO APARTMENT	420	studio	2010	NSW	Darlinghurst	3.6	264	3	5/3/2026, 6:20:58 PM
A PLACE TO TRULY CALL HOME - SP...	580	house	4744	Qld	Moranbah	3.45	234	3	5/3/2026, 5:21:32 PM
It will go quick! Call us now	375	duplex/semi-d...	4285	Qld	Beaudesert	3.59	179	4	5/2/2026, 6:32:22 PM
Stylish Waterside Living	630	apartment	4817	Qld	Thuringowa C...	3.51	156	5	5/3/2026, 6:40:52 PM

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Screenshot 5. Ratings Table

/ratings/rentals/{id}

This endpoint requires a logged in user token to be accessed. GET method is used to retrieve a user's rating about specific rental using its ID, if it exists. The POST method is used to leave a rating by the user. New rating overwrites the old one. These endpoints are used on specific rental's page to show user's rating and form to leave a rating (Screenshot 3, Screenshot 6), provided user is logged in.

Located in the heart of Castle Hill, you'll enjoy close proximity to quality schools, parks, public transport and major shopping at Castle Towers.

This is an outstanding opportunity to secure a renovated home with a p... To exit full screen, press **F11** ...t-after pockets!

Disclaimer: The above information has been gathered from sources that we believe are reliable. However, we cannot guarantee the accuracy of this information and nor do we accept responsibility for its accuracy. Any interested parties should rely on their own enquiries and judgment to determine the accuracy of this information for their own purposes. Images are for illustrative and design purposes only and do not represent the final product or finishes.

Rental info:

- Rent: \$980
- Property Type: house
- Locality: Castle Hill
- Postcode: 2154
- State: NSW
- Street Address: 9 Bowman Avenue
- Suburb: Castle Hill
- Bathrooms: 1
- Bedrooms: 4
- Parking spaces:
- Agency: Manor Real Estate
- Amenities: Balcony, Garage: 1, Open Spaces: 3, Outdoor Entertaining Area, Remote Garage, Secure Parking, Shed, Swimming Pool - Inground, Built-in Wardrobes, Floorboards, Split-system Air Conditioning, Toilets: 2
- Rating: 3.27 Stars (11)

Not rated by you yet
Rate this property:

Location



Screenshot 6. Rental Page without user rating

/user/register

POST endpoint used for registering new user by supplying email and password. The app provides user with a form to input an email, password (with confirmation), and displays error messages on page if there are any.

Rental Finder Home Rentals Ratings Login

Signup

Request body incomplete, both email and password are required

Enter email Enter password Confirm password

Signup

Already have an account? [Go to Login](#)

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Screenshot 7. Signup Page

</user/login>

POST endpoint used for logging in an existing user. The app provides user with a form to input an email and password , and displays error messages on page if there are any.

Rental Finder Home Rentals Ratings Login

Login

Log in to view and leave rental ratings

Enter email Enter password

Login

Don't have an account? [Go to Signup](#)

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Screenshot 8. Login Page

Modules Used

React Router

Module used to enable Single-Page Application functionality

<https://reactrouter.com/>

AG Grid

Module used to render a table containing rental and rating data

<https://www.ag-grid.com/react-data-grid/>

React Bootstrap

Used for component styling, layout and spacing

<https://react-bootstrap.netlify.app>

Pigeon Maps

Used to display individual property location on the map

<https://pigeon-maps.js.org/docs/>

React Toastify

Used to pop up a notification when user successfully logins, signups, logouts or leaves a rating

<https://pigeon-maps.js.org/docs/>

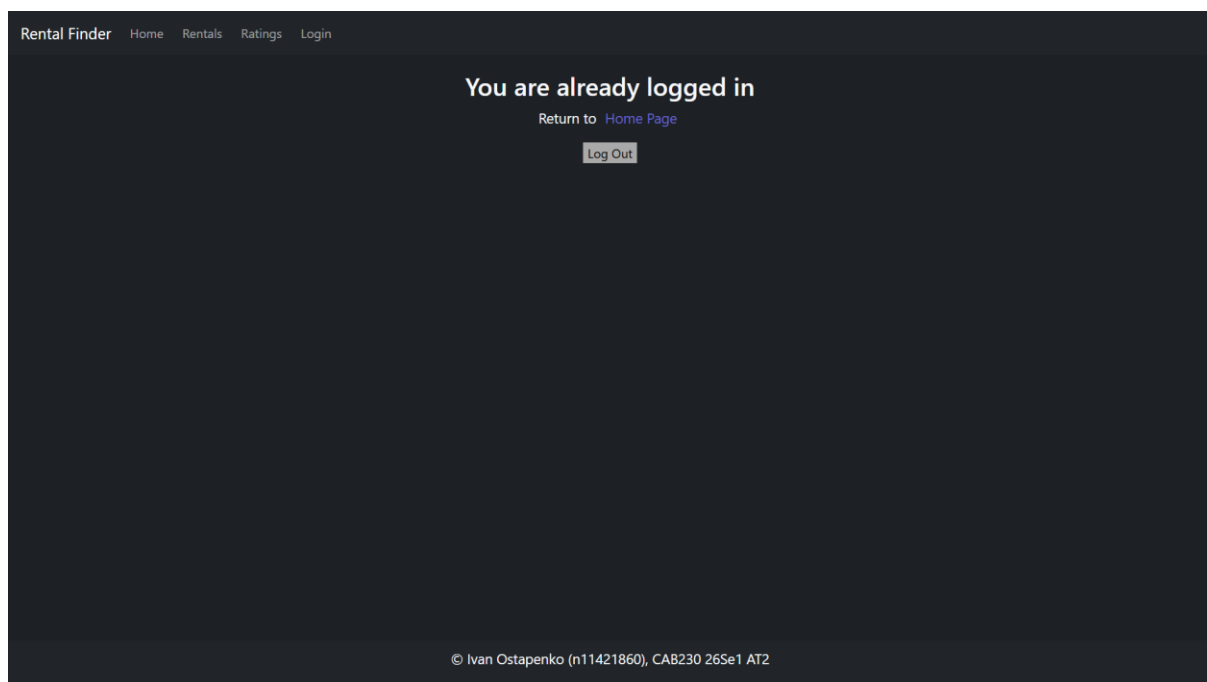
Application Design

Navigation and Layout

The application renders Navbar component at the top of the screen on every page, so that users may easily traverse between whichever pages they desire. Alternative, vertical or collapsable navbars were considered, however, the website's UI is pretty simplistic, and always-visible horizontal navbar looked nicer.

The login page has a link to the signup page and vice versa, to allow for easy-to-understand navigation. Getting to the individual rental page is possible only through tables or directly using URL, which may be a limitation, due to possibly large quantities of rentals in both tables.

Application also has a protected and public routes, meaning only logged in users may access ratings table and data (protected), while only visitors may access login and signup pages (public), the logged in user navigates to the special page which offers to log out or come back to home page.



Screenshot 9. Logged In Page

Usability and Quality of Design

The layout of the application is mostly consistent. Padding, gaps, font sizes, UI coloring is mostly consistent across the board, spacing between the elements of the UI provides clarity. The navbar at the top is clearly defined and provides user with the means to comfortably navigate across main pages. However, overall design is pretty lackluster. The positioning of the elements on the screen could be better, login and signup pages feel mostly empty and could probably use a different layout. The tables do not fill the screen completely, which is probably something that could be improved. The individual rental pages is the most unfinished page, the content layout is not visually intuitive to read. Overall, the design is clear and interactive enough to be usable, but not particularly well organized and loses in user experience to modern websites.

For rental search, min and max values were considered and enforced, with appropriate steps for numerical values. For values that are minimum and maximum (e.g. bathrooms, rating), value of max

cannot be lower than value of min. Sort Order can only work when Sort By is provided in the endpoint request, so in UI Sort Order can only be set after Sort By, and if Sort By is cleared it automatically clears Sort Order. Also a “Reset” button was added to the search, to clear all search fields and reset search, though these actions could probably be broken down into two different buttons.

Efforts were made to make navigation intuitive. Predicting potential user behavior and how they would traverse the website, relevant links were made (login, signup and logout pages). When accessing ratings while not being logged in, user is redirected to the login page. Text on the login page notifies user that ratings may only be accessed after login. The search feature on rentals table page could be positioned more horizontally, there is a lot of unoccupied place, as well as some of the search options could be shown at all times (like state, rent and postcode).

The UI style is simple, with overall dark theme, and with good color balance. Buttons, inputs, navbar, tables and hero image were all made with minimalistic design. Vibrant colors have been avoided as they would be out of place among existing palette (except for error messages during login and signup). Overall, although simple, color design appears to follow web design conventions reasonably well.

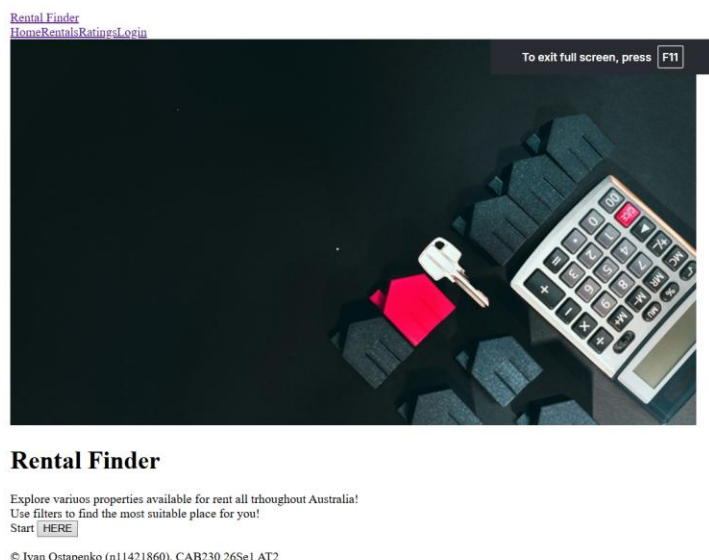
Accessibility

Priority 1 reflection:

The website uses only one non-text element, which is a hero image on the main page, and it does have a text alternative.

No information in this application is conveyed only through colour.

After Removing style sheets, the application elements are still visible and readable.



Screenshot 10. Application Without CSS

Text mostly dynamically updates when values are changed. The only exception that was observed was individual rental page, where leaving a rating does not change the text displayed until the page is manually reloaded. This is a bug that I failed to solve in time.

Screen flickering might be possible when scrolling through the tables when rows are being updated, or from UI elements positions changing if rapidly resizing window.

The language is mostly simple and unambiguous in the application, though some search parameters behaviour may be unclear to some users.

The tables content is clearly differentiated, although some field, like “rating” “total rating” “your ratings” may be confusing.

Inclusive design and cultural accessibility:

The Acknowledgement of Traditional Owners, though not necessary, may be included in the website structure. That is relatively easy to accomplish, for example putting it in the website footer.

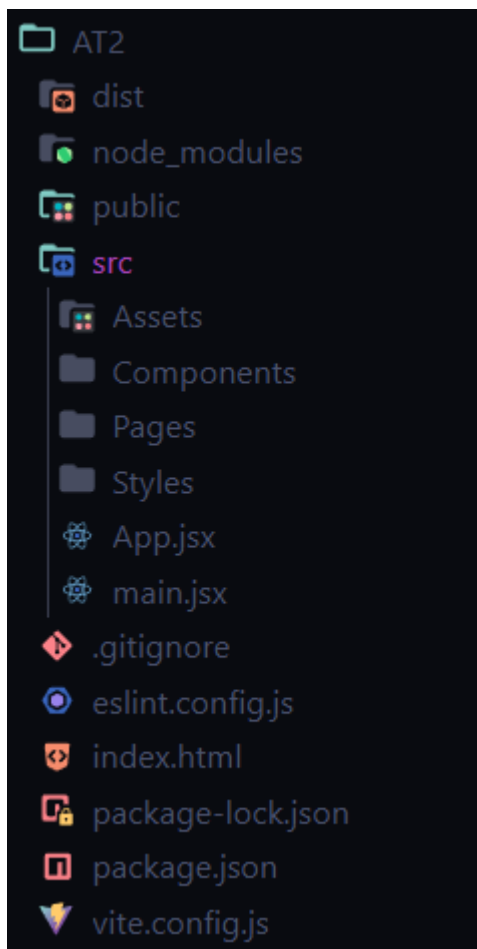
While it is unlikely that current design may affect Aboriginal and Torres Strait Islander users, the content of some of the rental posting definitely can and should be filtered accordingly.

Technical Description

Architecture

The *Screenshot 11* shows the application file structure. The src directory is broken down into multiple directories, based on their role in the application:

- **Assets** – contains assets, such as hero image (hat is the only asset in this application)
- **Components** - .jsx files, base component which are being used by pages and for layout.
- **Pages** - .jsx files, main pages which utilize components and are called by the browser-router
- **Styles** – .css files for the whole document and some specific pages or components
- **App.jsx** – a home page
- **main.jsx** – renders root component, router and ToastContainer (for notifications)

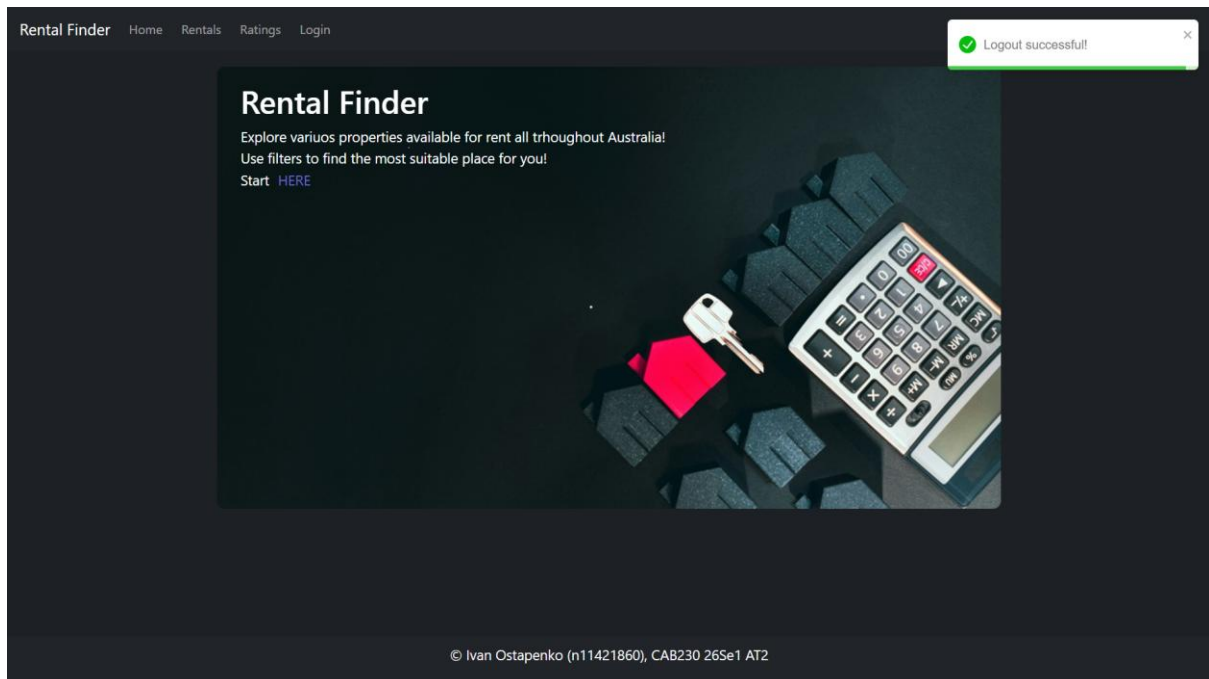


Screenshot 11. Application Structure

Test plan

Test	Expected result	Result	Screenshots
Navbar	Navbar is present on all pages	PASS	1-3, 5, 7-10
Rental Search	Search form is working and searching works	PASS	2
Map	Map is loaded and marker correctly positioned	PASS	4

Rating Rental Page	Rating is shown on rental page. If no rating is set display text stating so. Rating can be set on rental page.	PASS	3, 6
Toastify	Success message is displayed upon login/logout/signup	PASS	12
Login/signup error	Display error message during login/signup	PASS	7
Login/signup links	Have a link to navigate between login/signup	PASS	7, 8
Infinite row model	Infinite row model work in Rental and Rating tables	PASS	1, 5
Rating update	The rating value is updated when submitting the rating form	FAIL	
Visitor rating	Upon trying to access ratings, visitors are redirected to login page	PASS	8
Log out	When logged in user goes to login page, load logout page. On logout clear token	PASS	9
Rental Page	On row click in Rental or Rating table, redirect to Individual Rental page	PASS	1, 5: 3



Screenshot 12. Success message upon action

Difficulties / Exclusions / unresolved & persistent errors /

Roadblocks:

- Table search- it was challenging to create a search form and manage its layout in a graceful way, while including all possible functionality. Creating the form and its functionality as a

component separate from the page itself helped break down and clearly understand exactly what I need to do to implement it.

- UI and navbar design – I did not import bootstrap CSS and spent a lot of time trying to figure out how to make the application look generally nice
- Using flexbox for component layout – the <div>s were challenging to set up and nest in a way that would look nice and consistent across all pages
- Successful login/signup message – login and signup looked clunky, did not rerender the page properly and were unintuitive. I solved it by navigating to a different relevant page on successful submission and using React Toastify for notifications

Unfinished:

- Individual Rental page – multiple aspects of this page are bad and could be improved:
 - Layout – the text is positioned in a way that makes it clunky and difficult to distinguish which section they belong to. The rating text and forms seems out of place
 - Map – the map component is also positioned in a strange way, and is too zoomed out. I did not manage for the map to accept coordinates for API and center on the property while keeping the Marker positioned correctly, so had to settle with the whole Australia being on the map
 - Rating update – when submitting new rating, the page does not update shown data until reloaded.
- setloading – no pages use setLoading, so they do not have a substitute for when content takes too long to load

Bugs:

- Error handling – The application lacks proper error handling. For example, if getting error 404, there is no way to return to home page (*Screenshot 13*). A lot of other error handling is probably not handled
- Async/await – I believe I have missed a lot of places where Async/await should have been used

Unexpected Application Error!

404 Not Found

Hey developer 🙌

To exit full screen, press **F11**

You can provide a way better UX than this when your app throws errors by providing your own `ErrorBoundary` or `errorElement` prop on your route.

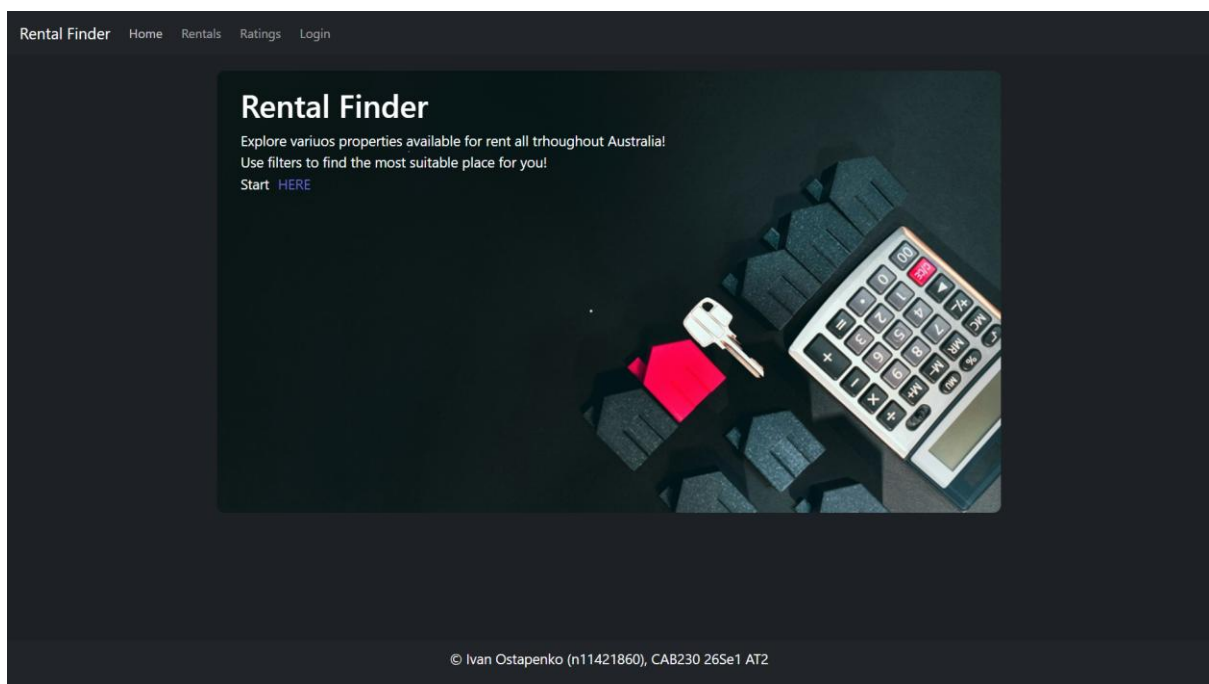
Screenshot 13. Error 404 page

Extensions

- UI Improvements
- Individual rental Page rework
- Map fix
- Error handling
- Loading handling

User guide

Upon opening the application, you land on the main page:

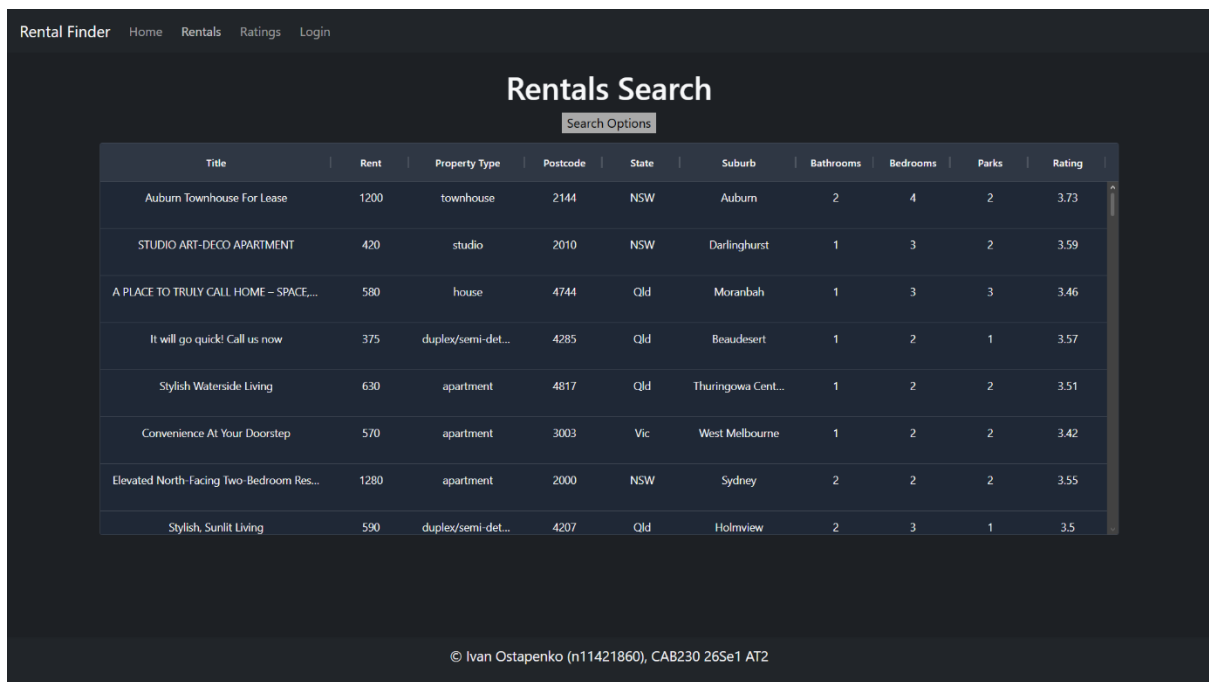


From here you can use:

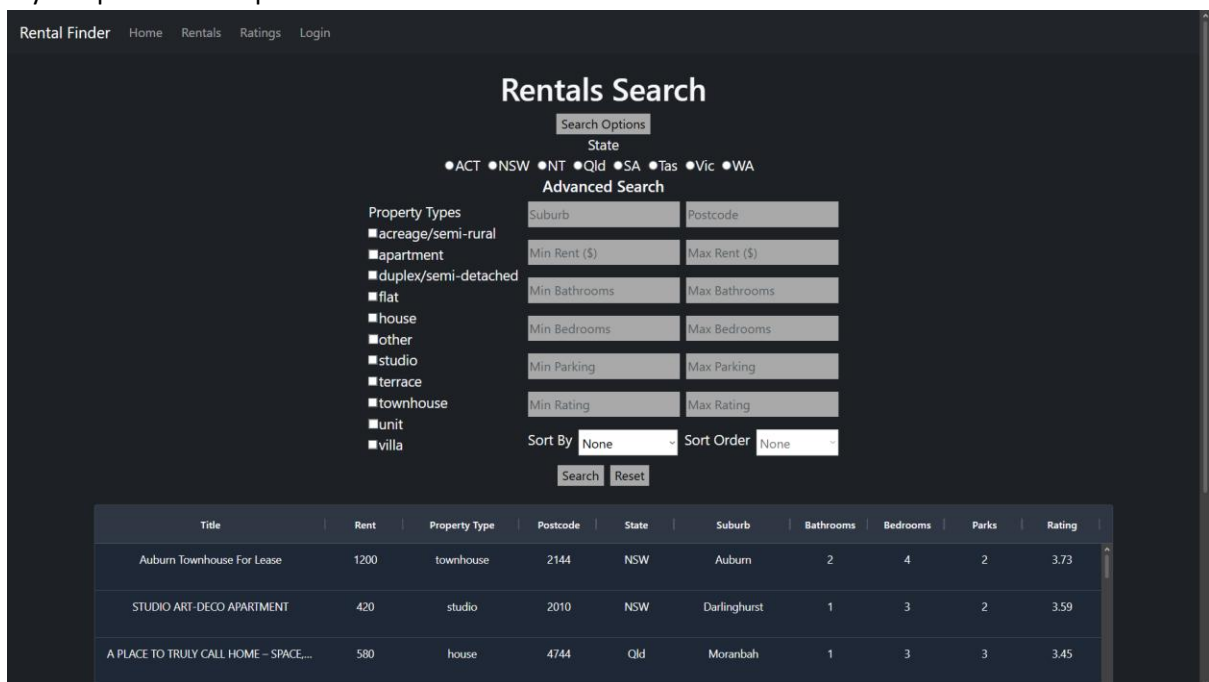
- Navbar (present across all pages):
 - "Rental Finder" or "Home" leads to home page
 - "Rentals" lead to rentals table with search
 - "Ratings" lead to a table with your ranked properties (if you are not logged in you are redirected to login page)
 - "Login" leads you to login page where you can login or signup (if you are logged in you are redirected to the logout page)
- "Here" on Hero image – leads you to rentals table with search

If you go to the rental table, you can see Rentals Search Table:

- If you click on Search Options, you will see various search parameters that you can use to filter the table
- If you click on any row: you are redirected to the individual rental page



If you open Search Options:



Possible search parameters (max parameters cannot be lower than min:

- State – only one state may be selected
- Property Types – multiple property types may be selected
- Suburb
- Postcode
- Min/max rent
- Min/max bathrooms
- Min/max bedrooms
- Min/max rating (max 5)
- Sort By – sort by any field, event total ratings, latitude, longitude

- **Sort Order** – ascending or descending, ascending by default, can only be set with Sort By
- **Search** – search with entered filters
- **Reset** – delete all entered search filters from the UI and reset the table

If you click on row:

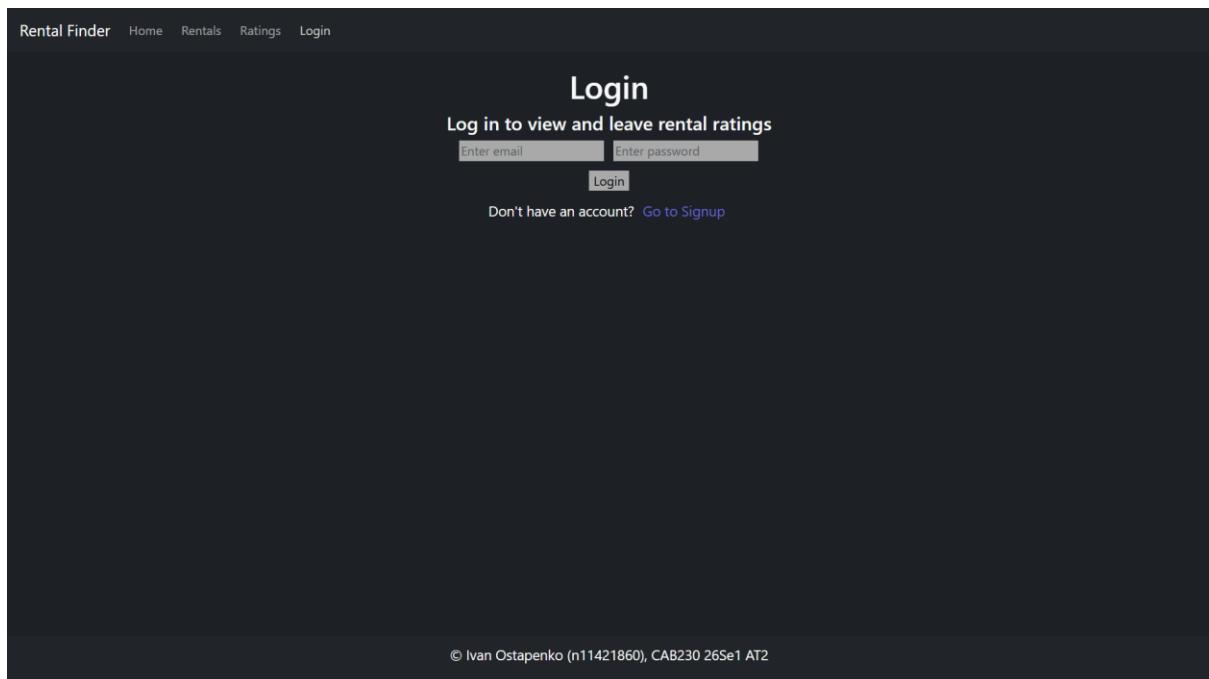
You can see rental title, description, and all data, as well as map with marker placed.

Here you can:

- Not see your rating if you are not logged in, and navigate to login page if you click the respective button

When you are logged in you can see your rating if you left any, and update rating:

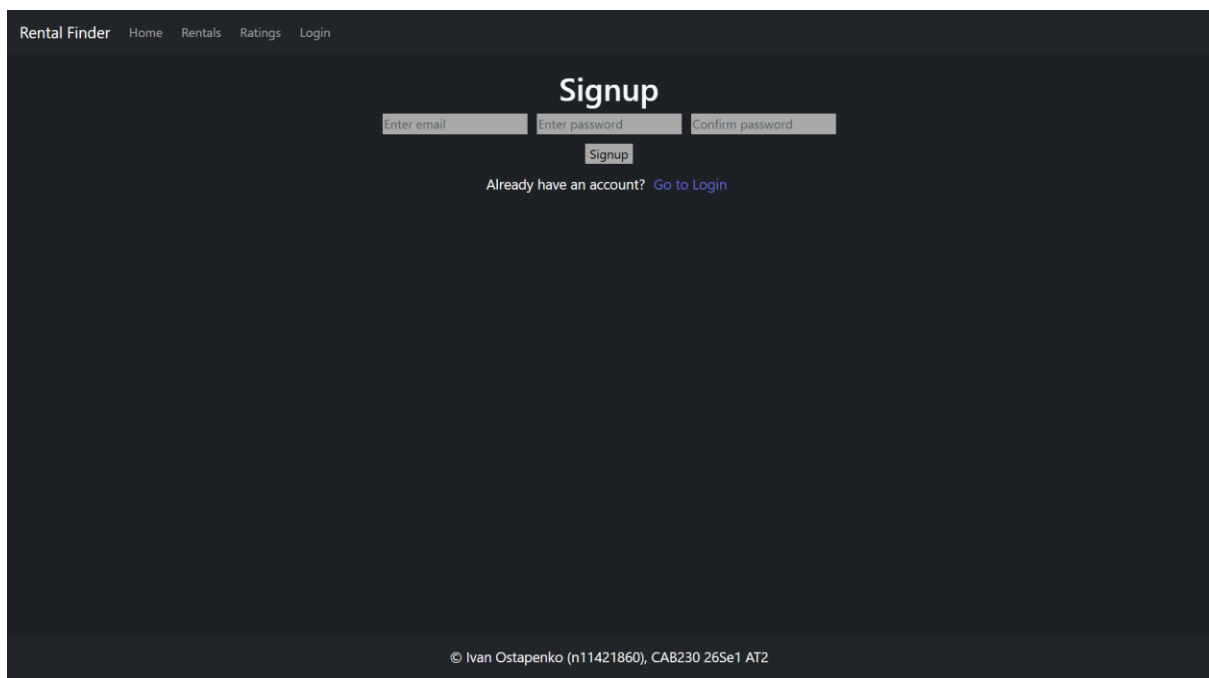
If you try opening Ratings while not logged in, you will be redirected to login page:



The screenshot shows the 'Login' page of the 'Rental Finder' application. The navigation bar at the top includes 'Rental Finder', 'Home', 'Rentals', 'Ratings', and 'Login'. The main heading is 'Login', followed by the instruction 'Log in to view and leave rental ratings'. There are two input fields: 'Enter email' and 'Enter password'. Below these is a 'Login' button. A link 'Go to Signup' is provided for users who don't have an account. The footer contains the copyright notice: '© Ivan Ostapenko (n11421860), CAB230 26Se1 AT2'.

Provide you email and password to login.

Or press the link to go to signup page:



The screenshot shows the 'Signup' page of the 'Rental Finder' application. The navigation bar at the top includes 'Rental Finder', 'Home', 'Rentals', 'Ratings', and 'Login'. The main heading is 'Signup'. There are three input fields: 'Enter email', 'Enter password', and 'Confirm password'. Below these is a 'Signup' button. A link 'Go to Login' is provided for users who already have an account. The footer contains the copyright notice: '© Ivan Ostapenko (n11421860), CAB230 26Se1 AT2'.

Here you enter your email, password and confirm password. Once your signed up, you will see a notification and need to login into your new account

Once logged in, you will see a notification and be redirected to your ratings table:

[Rental Finder](#) [Home](#) [Rentals](#) [Ratings](#) [Login](#)

Login successful!

Your Ratings

Title	Rent	Property Type	Postcode	State	Suburb	Rating	Total Ratings	Your Ra...	Rated
Auburn Townhouse For Lease	1200	townhouse	2144	NSW	Auburn	3.74	580	3	5/3/2026, 4:27:56 PM
A PLACE TO TRULY CALL HOME – SP...	580	house	4744	Qld	Moranbah	3.45	258	2	5/3/2026, 4:07:02 PM
Modern 4 Bedroom Townhouse Livin...	2100	townhouse	2070	NSW	Lindfield	3.7	54	3	5/3/2026, 4:35:39 PM

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Here you can see the properties that you rated, rating itself, and datetime when rating was submitted. When clicking on row, you will be redirected on individual rental page, like in rental table.

If you press login while logged in, you will go to logout page:

[Rental Finder](#) [Home](#) [Rentals](#) [Ratings](#) [Login](#)

You are already logged in

Return to [Home Page](#)

Log Out

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From here you can go back to home page, or logout.

Once you log out, you will see a notification and be redirected to the main page.

That concludes the User Guide.